



A Level • Cambridge (CIE) • Chemistry

28 mins

15 questions

Multiple Choice Practice

Topical Practice MCQ

4. States of matter

Carefully selected past paper questions by topic.

13 In this question you should assume that the gas formed behaves as an ideal gas.

A 1.7 g sample of Mg reacts with 50.0 cm³ of 2.2 mol dm⁻³ HCl at 303 K and 110 400 Pa.

Which volume of gas is produced, measured under these conditions?

- A** 1.3 dm³ **B** 1.6 dm³ **C** 2.5 dm³ **D** 5.0 dm³

7 Which assumptions are made about ideal gases?

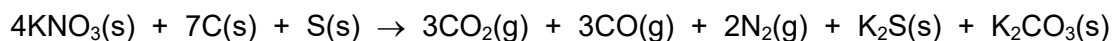
- 1 Ideal gases contain molecules with no mass.
- 2 Ideal gases contain molecules with no volume.
- 3 Ideal gases have no intermolecular forces.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Question 3

9701_s22_qp_11 • Q7

- 7 'Black powder' is a mixture of potassium nitrate, carbon and sulfur. The mixture reacts as shown.



A sealed tube containing black powder has a volume of 10.0 cm^3 . When all of the black powder reacts, the reaction causes a pressure of $2 \times 10^6\text{ Pa}$ and a temperature of 2500 K .

The volume of the K_2CO_3 and K_2S produced can be ignored.

How many moles of KNO_3 are contained in the sealed tube?

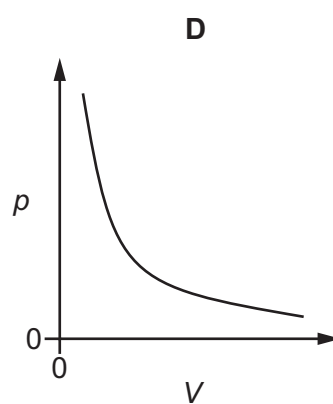
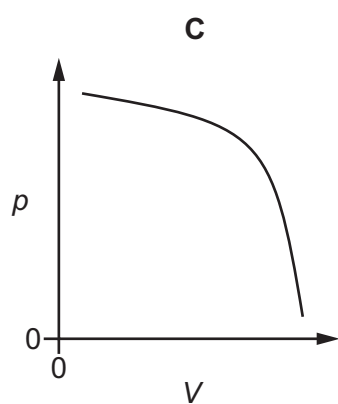
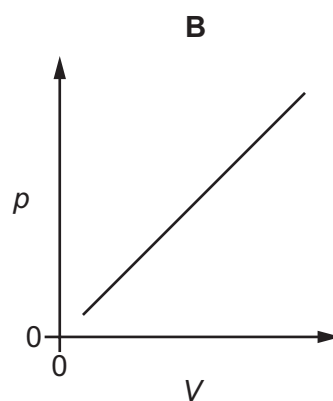
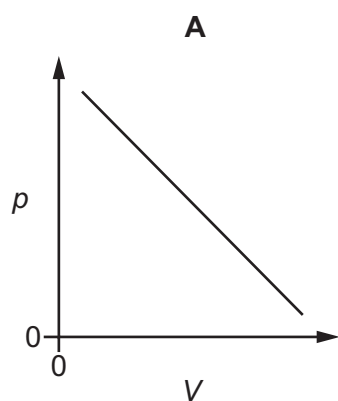
- A** 4.81×10^{-4} **B** 9.63×10^{-4} **C** 1.93×10^{-3} **D** 9.63×10^{-1}

- 7 Which gas will behave **least** like an ideal gas at 150 °C and 101 kPa?
- A ammonia
 - B fluorine
 - C krypton
 - D steam

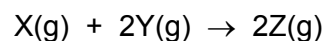
Question 5

9701_w22_qp_12 • Q15

- 15 Which graph represents the variation of pressure p and volume V of a sample of an ideal gas at constant temperature?



- 6 X, Y and Z are all gases that behave ideally and react according to the equation shown.



When 3.0 mol of X and 3.0 mol of Y are placed inside a container with a volume of 1.0 dm^3 , they react to form the maximum amount of Z.

The final temperature of the reaction vessel is 120°C .

What is the final pressure inside the reaction vessel?

- A $4.49 \times 10^6 \text{ Pa}$
- B $9.80 \times 10^6 \text{ Pa}$
- C $1.47 \times 10^7 \text{ Pa}$
- D $1.96 \times 10^7 \text{ Pa}$

- 6** Which solid contains more than one type of bonding?
- A** iodine
 - B** silicon dioxide
 - C** sodium chloride
 - D** zinc

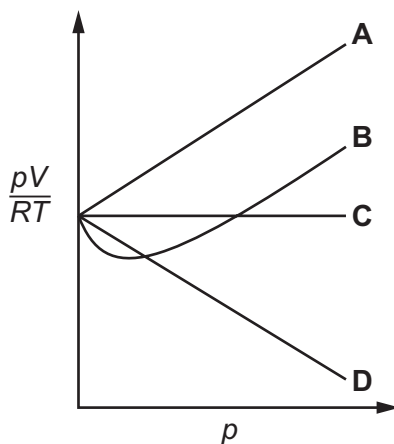
Question 8

9701_w17_qp_12 • Q5

- 5 A sample of an ideal gas is contained at a constant temperature of 300 K in a gas syringe.

The pressure is increased and a graph of $\frac{pV}{RT}$ against pressure is plotted.

Which graph correctly represents the results?



- 8** Solid carbon dioxide, CO_2 , is similar to solid iodine, I_2 , in its structure and properties. Carbon is in Group 14. Silica, SiO_2 , is a Group 14 compound.

Which statement about solid CO_2 and solid SiO_2 is correct?

- A** Both solids exist in a lattice structure.
- B** Both solids have a simple molecular structure.
- C** Both solids have atoms joined by single covalent bonds.
- D** Both solids change spontaneously to gas at s.t.p.

- 8 In an experiment, a sample of a pure gas is put into a gas syringe at a temperature of 300 K and pressure of 16 kPa. The gas is compressed until the volume occupied by the gas is halved.

After compression, the temperature of the gas in the syringe is 375 K and the pressure is 40 kPa.

Which statement is correct?

- A Intermolecular forces between the gas molecules are significant.
- B It is possible to calculate the number of moles of gas present using these data alone.
- C The gas is behaving ideally.
- D The pressures used are too high for ideal gas behaviour.

- 5 Some car paints contain small flakes of silica, SiO_2 .

In the structure of solid SiO_2

- each silicon atom is bonded to **x** oxygen atoms,
- each oxygen atom is bonded to **y** silicon atoms,
- each bond is a **z** type bond.

What is the correct combination of **x**, **y** and **z** in these statements?

	x	y	z
A	2	1	covalent
B	2	1	ionic
C	4	2	covalent
D	4	2	ionic

2 *Use of the Data Booklet is relevant to this question.*

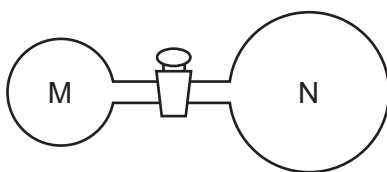
Iodine is a black, shiny, non-metallic solid and a member of Group VII. It sublimes easily on heating to give a purple vapour.

A sample of iodine vapour of mass 6.35 g has a volume of 1.247 dm³ when maintained at constant temperature and a pressure of 1.00×10^5 Pa.

If iodine vapour acts as an ideal gas, what is the temperature of the iodine vapour?

- A** 300 K **B** 600 K **C** 300 000 K **D** 600 000 K

- 4 Two glass vessels M and N are connected by a closed valve.

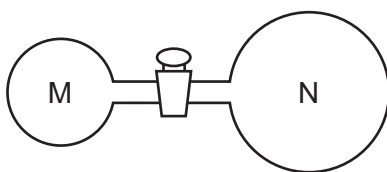


M contains helium at 20°C at a pressure of $1 \times 10^5 \text{ Pa}$. N has been evacuated, and has three times the volume of M. In an experiment, the valve is opened and the temperature of the whole apparatus is raised to 100°C .

What is the final pressure in the system?

- A $3.18 \times 10^4 \text{ Pa}$
- B $4.24 \times 10^4 \text{ Pa}$
- C $1.25 \times 10^5 \text{ Pa}$
- D $5.09 \times 10^5 \text{ Pa}$

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Answer Key

No.	Paper	Topic	Answer
1	9701_s23_qp_11	4. States of matter	A
2	9701_s23_qp_13	4. States of matter	D
3	9701_s22_qp_11	4. States of matter	A
4	9701_s22_qp_12	4. States of matter	D
5	9701_w22_qp_12	4. States of matter	D
6	9701_s19_qp_12	4. States of matter	C
7	9701_s18_qp_11	4. States of matter	A
8	9701_w17_qp_12	4. States of matter	C
9	9701_m16_qp_12	4. States of matter	A
10	9701_w16_qp_12	4. States of matter	C
11	9701_w15_qp_13	4. States of matter	C
12	9701_s14_qp_11	4. States of matter	B
13	9701_w14_qp_11	4. States of matter	A
14	9701_w14_qp_12	4. States of matter	A
15	9701_w12_qp_11	4. States of matter	C